

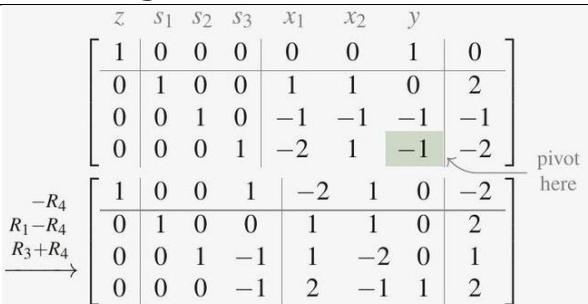
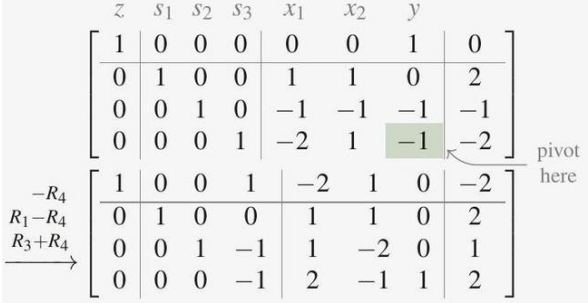
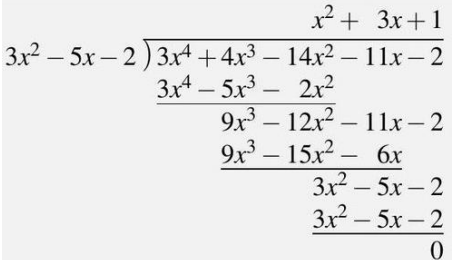
johnston-linear-matrix-algebra: residuals

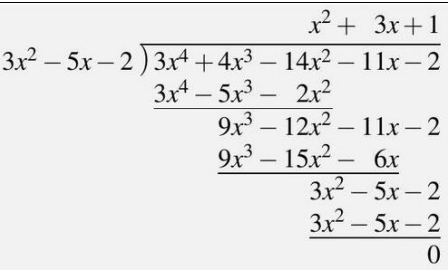
equations measured 2026-09-13T12:05:45Z against the model of 2026-09-13

Corrected (6)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra-EQ0006 (was)	22	0.141	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$
johnston-linear-matrix-algebra-EQ0006 (now)	22	—	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$
basis: inferred / ink					
johnston-linear-matrix-algebra-EQ0014 (was)	27	0.071	$\left. \begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array} \right\} \text{ property (property (a)) }$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$	$\begin{array}{lrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ & & \mathbf{w}) \text{ } & =-\mathbf{v}-3\mathbf{x}+3\mathbf{w} \\ & & \mathbf{w}) \text{ } & =-3\mathbf{v}+\mathbf{w} \\ & & \mathbf{w}) \text{ } & =\frac{1}{4}(\mathbf{w}-3\mathbf{v}). \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra-EQ0014 (now)	27	—	$\begin{array}{l} \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{x} + \mathbf{y} \end{array} = \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{x} \end{array} + \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{y} \end{array} \quad (\text{property (b)}) \\ \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{x} + \mathbf{y} \end{array} = \begin{array}{l} \mathbf{x} \end{array} \cdot \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} + \begin{array}{l} \mathbf{y} \end{array} \cdot \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \quad (\text{property (a)}) \\ \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{x} + \mathbf{y} \end{array} = \begin{array}{l} \mathbf{x} \end{array} \cdot \mathbf{v} + \begin{array}{l} \mathbf{x} \end{array} \cdot \mathbf{w} + \begin{array}{l} \mathbf{y} \end{array} \cdot \mathbf{v} + \begin{array}{l} \mathbf{y} \end{array} \cdot \mathbf{w} \quad (\text{property (b)}) \\ \begin{array}{l} \mathbf{v} + \mathbf{w} \end{array} \cdot \begin{array}{l} \mathbf{x} + \mathbf{y} \end{array} = \mathbf{v} \cdot \mathbf{x} + \mathbf{w} \cdot \mathbf{x} + \mathbf{v} \cdot \mathbf{y} + \mathbf{w} \cdot \mathbf{y}. \quad (\text{property (a)}) \end{array}$	$\begin{array}{l} (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{x} + \mathbf{y}) = (\mathbf{v} + \mathbf{w}) \cdot \mathbf{x} + (\mathbf{v} + \mathbf{w}) \cdot \mathbf{y} \quad (\text{property (b)}) \\ = \mathbf{x} \cdot (\mathbf{v} + \mathbf{w}) + \mathbf{y} \cdot (\mathbf{v} + \mathbf{w}) \quad (\text{property (a)}) \\ = \mathbf{x} \cdot \mathbf{v} + \mathbf{x} \cdot \mathbf{w} + \mathbf{y} \cdot \mathbf{v} + \mathbf{y} \cdot \mathbf{w} \quad (\text{property (b)}) \\ = \mathbf{v} \cdot \mathbf{x} + \mathbf{w} \cdot \mathbf{x} + \mathbf{v} \cdot \mathbf{y} + \mathbf{w} \cdot \mathbf{y}. \quad (\text{property (a)}) \end{array}$	$\begin{array}{l} (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{x} + \mathbf{y}) = (\mathbf{v} + \mathbf{w}) \cdot \mathbf{x} + (\mathbf{v} + \mathbf{w}) \cdot \mathbf{y} \quad (\text{property (b)}) \\ = \mathbf{x} \cdot (\mathbf{v} + \mathbf{w}) + \mathbf{y} \cdot (\mathbf{v} + \mathbf{w}) \quad (\text{property (a)}) \\ = \mathbf{x} \cdot \mathbf{v} + \mathbf{x} \cdot \mathbf{w} + \mathbf{y} \cdot \mathbf{v} + \mathbf{y} \cdot \mathbf{w} \quad (\text{property (b)}) \\ = \mathbf{v} \cdot \mathbf{x} + \mathbf{w} \cdot \mathbf{x} + \mathbf{v} \cdot \mathbf{y} + \mathbf{w} \cdot \mathbf{y}. \quad (\text{property (a)}) \end{array}$
basis: inferred / ink					
johnston-linear-matrix-algebra-EQ0063 (was)	45	■0.136	$\begin{aligned} A B &= \left[\begin{array}{cc} I_3 & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] \\ &= \left[\begin{array}{cc} I_3 & D + I_3 \\ 0 & O \end{array} \right] \left[\begin{array}{cc} O & O + I_3 \\ O & D + C \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right]. \end{aligned}$	$AB = \left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} I_3 & D + I_3 \\ O & D + C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right].$	$AB = \left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} I_3 & D + I_3 \\ O & D + C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right].$
johnston-linear-matrix-algebra-EQ0063 (now)	45	—	$\begin{aligned} &\Rightarrow 4\mathbf{x} = -3\mathbf{v} + \mathbf{w} \quad (\text{add } 3\mathbf{x}, \text{ subtract } 2\mathbf{v}) \\ &\Rightarrow \mathbf{x} = \frac{1}{4}(\mathbf{w} - 3\mathbf{v}). \quad (\text{divide both sides by } 4) \end{aligned}$	$\begin{aligned} &\Rightarrow 4\mathbf{x} = -3\mathbf{v} + \mathbf{w} \quad (\text{add } 3\mathbf{x}, \text{ subtract } 2\mathbf{v}) \\ &\Rightarrow \mathbf{x} = \frac{1}{4}(\mathbf{w} - 3\mathbf{v}). \quad (\text{divide both sides by } 4) \end{aligned}$	$AB = \left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} I_3 & D + I_3 \\ O & D + C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right].$
basis: inferred / ink					
johnston-linear-matrix-algebra-EQ0486 (was)	196	■0.011	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$
johnston-linear-matrix-algebra-EQ0486 (now)	196	—	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$	$\begin{array}{l} \mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - \mathbf{z} - 1 \\ \mathbf{v} - 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y} + 2\mathbf{z} + 2 \\ -\mathbf{v} + \mathbf{w} + \mathbf{x} + \mathbf{y} - 3\mathbf{z} + 3 \\ 2\mathbf{v} + \mathbf{w} - 2\mathbf{y} + \mathbf{z} - 4 \end{array}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra- EQ0528 (was)	212	■ 0.001	$\begin{array}{r} \leftarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ 0 \\ 0 \\ 0 \end{array} \left \begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right \begin{array}{c} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right] \\ 2 \\ \text{pivot} \end{array} \end{array}$		
johnston-linear-matrix-algebra- EQ0528 (now)	212	—	$\begin{array}{l} \rightarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ R_4 + R_4 \end{array} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{array} \right] \curvearrowright \text{pivot here} \xrightarrow{\begin{array}{c} -R_4 \\ R_1 - R_4 \\ R_3 + R_4 \end{array}} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right] \end{array}$		
basis: inferred / ink					
johnston-linear-matrix-algebra- EQ1276 (was)	427	■ 0.022	$\left(3x^2 - 5x - 2 \right) \longdiv { \frac { x ^ { 2 } } { 3x^4 + 4x^3 - 14x^2 - 11x - 2 } , \frac { 3x^4 - 5x^3 - 2x^2 } { 9x^3 - 12x^2 - 11x - 2 } , \frac { 9x^3 - 15x^2 - 6x } { 3x^2 - 5x - 2 } , \frac { 3x^2 - 5x - 2 } { 0 } }$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

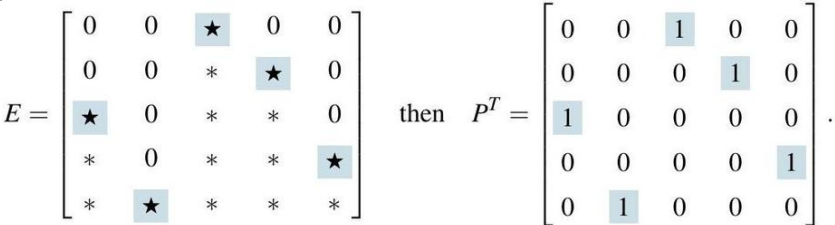
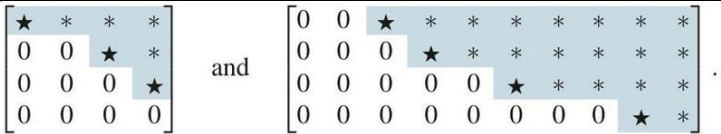
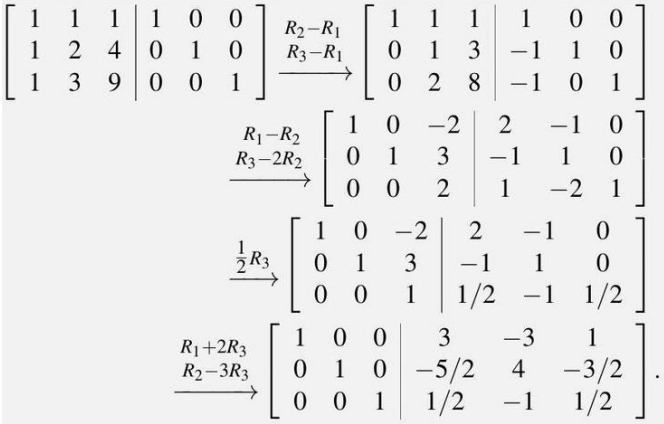
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-atlas E01276 (now)	427	—	$\begin{array}{r} x^2+3x+1\backslash \\ \backslash 3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}\backslash\backslash \\ \underline{3x^4-5x^3-2x^2}\backslash\backslash \\ 9x^3-12x^2-11x-2\backslash\backslash \underline{9x^3-15x^2-6x}\backslash\backslash \\ \backslash\backslash 3x^2-5x-2\backslash\backslash \underline{3x^2-5x-2}\backslash\backslash 0 \end{array}$	$3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}$ $\underline{3x^4-5x^3-2x^2}$ $9x^3-12x^2-11x-2$ $\underline{9x^3-15x^2-6x}$ $3x^2-5x-2$ $\underline{3x^2-5x-2}$ 0	
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-E00114 [conf 0.025]	63	0.025	<pre>\begin{figure} \[R^{\theta}\left(\mathbf{e}_{2}\right)=(-\sin (\theta), \cos (\theta)) \overbrace{\cos (\theta)^{-\sin (\theta)}}^{\mathbf{e}_{1}^{\mathbf{e}_{2}}} \times \text { - } \underbrace{\cos (\theta)}_{R^{\theta}\left(\mathbf{e}_{1}\right)=\left(\cos (\theta), \sin (\theta)\right)} \]</pre> <code>\captionsetup{labelformat=empty} \caption{Figure 1.19: (R^{θ}) rotates the standard basis vectors $(\mathbf{e}_1)$ and $(\mathbf{e}_2)$ to $(R^{\theta}\left(\mathbf{e}_1\right))=(\cos (\theta), \sin (\theta))$ and $(R^{\theta}\left(\mathbf{e}_2\right))=(-\sin (\theta), \cos (\theta))$, respectively.}</code>	(not rendered)	
johnston-linear-matrix-algebra-E00439	170	0.131	<pre>\begin{figure} \[\begin{aligned} {[A \mid I]} &= \left[\begin{array}{cccccccc cccc} * & * & * & * & * & * & * & * & 1 & 0 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 1 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 1 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 0 & 1 \end{array} \right] \\ \xrightarrow{\text{row-reduce}} & \left[\begin{array}{cccccccc cccc} \star & * & * & * & * & * & * & * & * & * & * & * \\ 0 & 0 & \star & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \star & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{array} \right] = [R \mid E] \end{aligned}</pre> <code>\leftarrow \text{null}\left(A^T\right)</code> <code>\operatorname{range}\left(A^T\right)</code> <code>\captionsetup{labelformat=empty} \caption{Figure 2.22: After we row-reduce $([A \mid I])$ to a row echelon form $([R \mid E])$, we can immediately read off bases of $(\operatorname{range}\left(A^T\right))$ and $(\operatorname{null}\left(A^T\right))$ from the rows of that matrix.}</code>	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ0884	318	■ 0.451 ● C +38	$\begin{aligned} \operatorname{rank}(A) &= \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \\ \det(A) &= \det(B) = 2, \quad \text{and} \quad p_A(\lambda) = p_B(\lambda) = -\lambda^3 + 3\lambda + 2. \end{aligned}$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$ $\det(A) = \det(B) = 2, \quad \text{and} \quad p_A(\lambda) = p_B(\lambda) = -\lambda^3 + 3\lambda + 2.$
johnston-linear-matrix-algebra EQ0793	293	■ 0.462 ● C +130	$\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{array} \right] \quad \text{has row echelon form} \quad \left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & R & \end{array} \right],$	$\left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & R & \end{array} \right],$	$\left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & R & \end{array} \right],$
johnston-linear-matrix-algebra EQ0040	39	■ 0.479 ● C +817	$\begin{aligned} A B &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra EQ0073	48	■ 0.493 ● C +396	$\begin{aligned} A B &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ0635 (0)	243	■0.508 ●C +27	E=\left[\begin{array}{ccccc} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{array}\right] \text{ then } P^T=\left[\begin{array}{ccccc} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right] .		
johnston-linear-matrix-algebra EQ0214	101	■0.533 ●C +146	\left[\begin{array}{llll} \star & \star & \star & \star \\ \star & \star & \star & \star \\ \star & \star & \star & \star \\ \star & \star & \star & \star \end{array}\right] \quad \text{and} \quad \left[\begin{array}{cccccccc} 0 & 0 & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & \star & \star \end{array}\right] .		
johnston-linear-matrix-algebra EQ0331 (0)	130	■0.564 ●C +20	\begin{array}{r} \left[\begin{array}{l} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array} \middle \begin{array}{l} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{l} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array} \middle \begin{array}{l} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{array} \right] \\ \xrightarrow{R_1-R_2} \left[\begin{array}{l} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array} \middle \begin{array}{l} 2 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & -2 & 1 \end{array} \right] \\ \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{l} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array} \middle \begin{array}{l} 2 & -1 & 0 \\ -1 & 1 & 0 \\ 1/2 & -1 & 1/2 \end{array} \right] \\ \xrightarrow[R_2-3R_3]{R_1+2R_3} \left[\begin{array}{l} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \middle \begin{array}{l} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array} \right] . \end{array}		
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0519	207	0.720 C +175	$\begin{gathered} \left[\begin{array}{cccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right] \\ 5/2 \text{ is smallest (and only) positive ratio} \end{gathered}$		
johnston-linear-matrix-algebra-EQ0210	99	0.827 C +26	$\begin{gathered} \left[\begin{array}{cccc cccc} 1 & 3 & -2 & 5 & 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 2 & 4 & 5 & 12 & 1 & 3 & -2 & 5 \end{array} \right] \\ \xrightarrow{R_2-R_1} \left[\begin{array}{cccc cccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 2 & 4 & 5 & 12 & 1 & 3 & -2 & 5 \end{array} \right] \\ \xrightarrow{R_3-2R_1} \left[\begin{array}{cccc cccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 0 & 2 & -6 & 4 & 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 & 0 & 0 & 3 & 6 \end{array} \right] \\ \xrightarrow{R_3+R_2} \left[\begin{array}{cccc cccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 0 & 2 & -6 & 4 & 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 & 0 & 0 & 3 & 6 \end{array} \right] \end{gathered}$		
johnston-linear-matrix-algebra-EQ1047	359	0.865 C +60	$\begin{aligned} & x_1 = \frac{\det \left(\begin{bmatrix} b_1 & a_{1,2} \\ b_2 & a_{2,2} \end{bmatrix} \right)}{\det \left(\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \right)} = \frac{b_1 a_{2,2} - a_{1,2} b_2}{a_{1,1} a_{2,2} - a_{1,2} a_{2,1}} \text{ and} \\ & x_2 = \frac{\det \left(\begin{bmatrix} a_{1,1} & b_1 \\ a_{2,1} & b_2 \end{bmatrix} \right)}{\det \left(\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \right)} = \frac{a_{1,1} b_2 - b_1 a_{2,1}}{a_{1,1} a_{2,2} - a_{1,2} a_{2,1}}. \end{aligned}$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					